



Driver Drowsiness Detection using CNN

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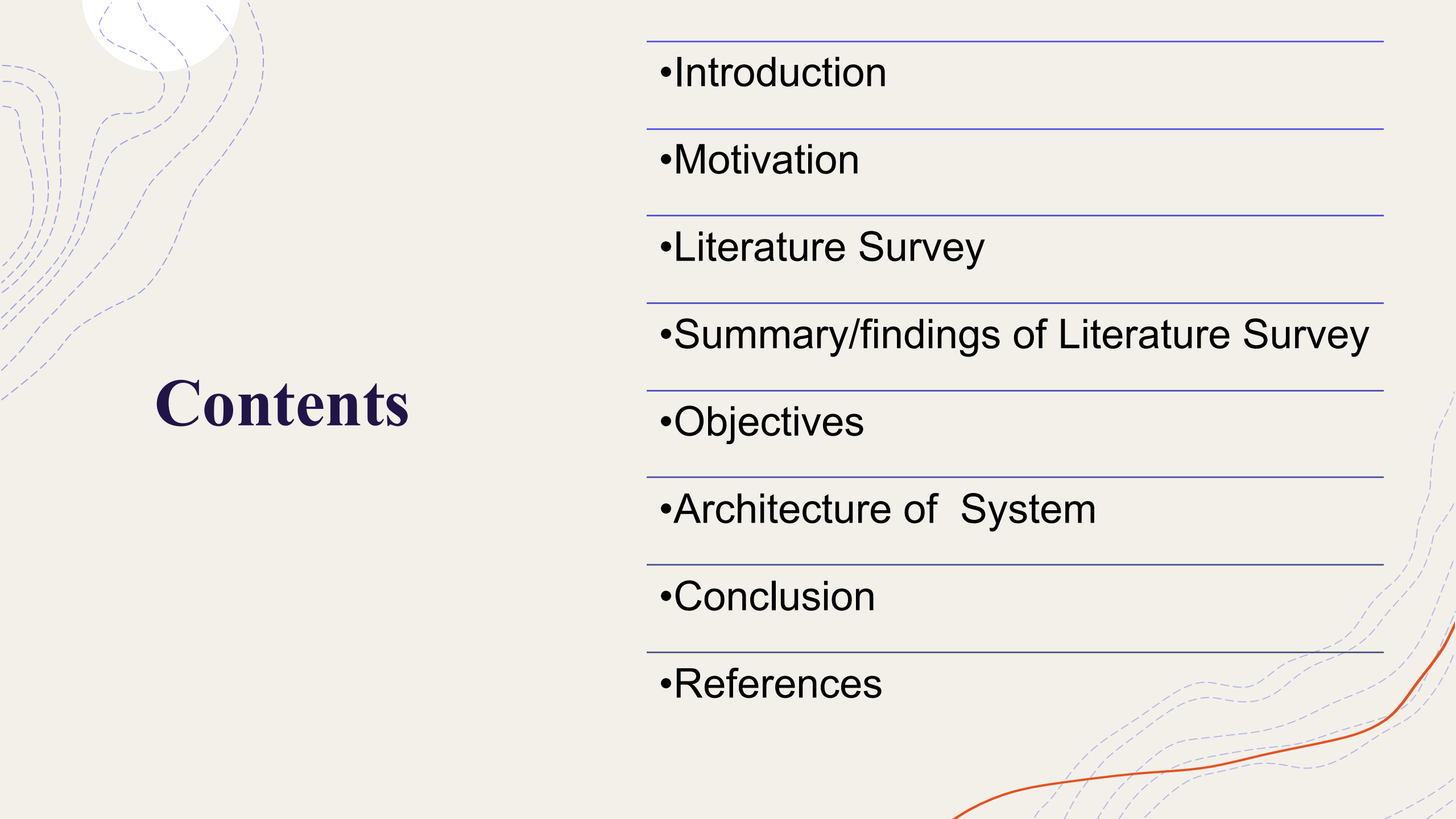
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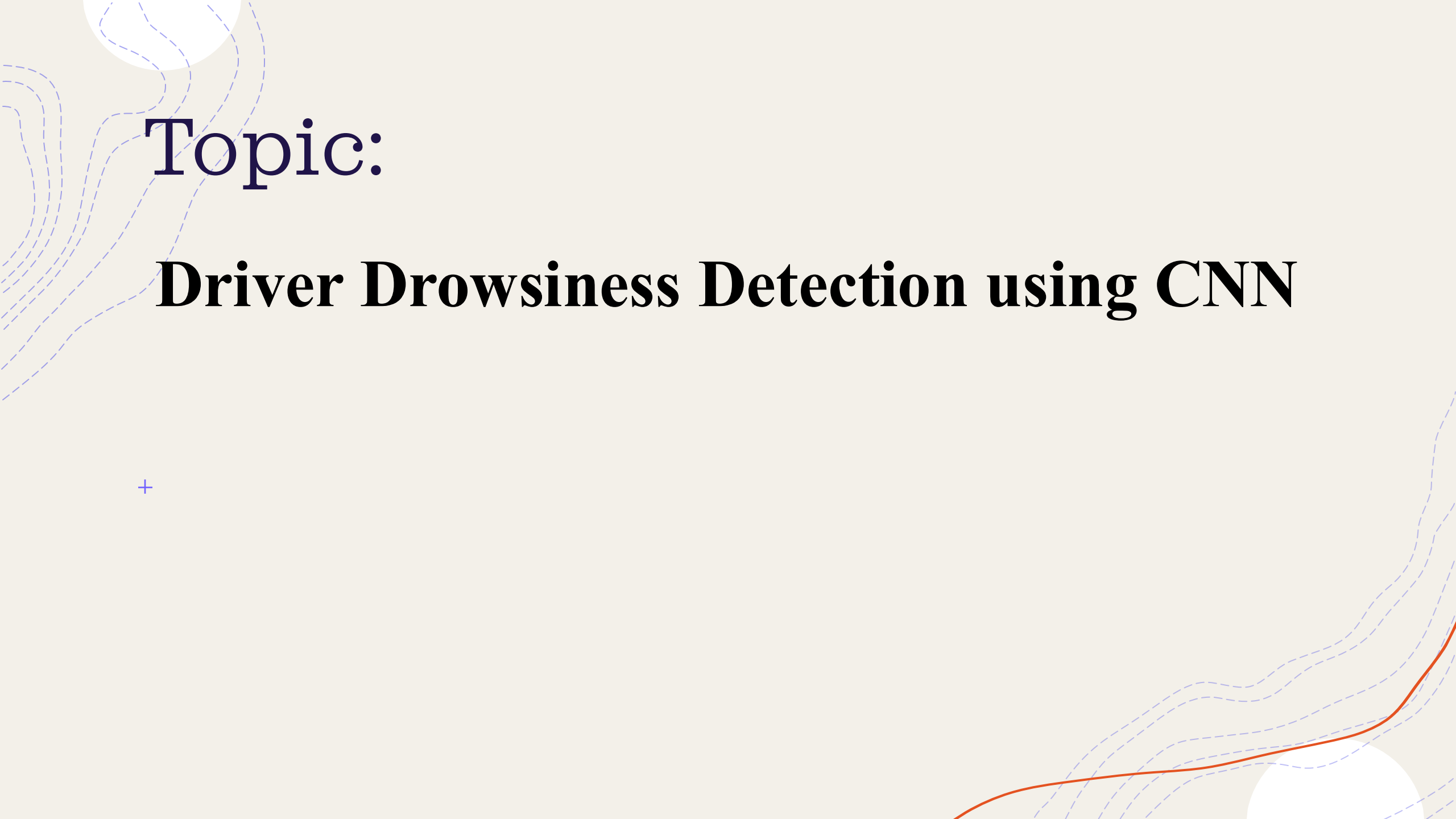
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Topic:

Driver Drowsiness Detection using CNN

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Introduction

+ Drowsiness and Road Safety

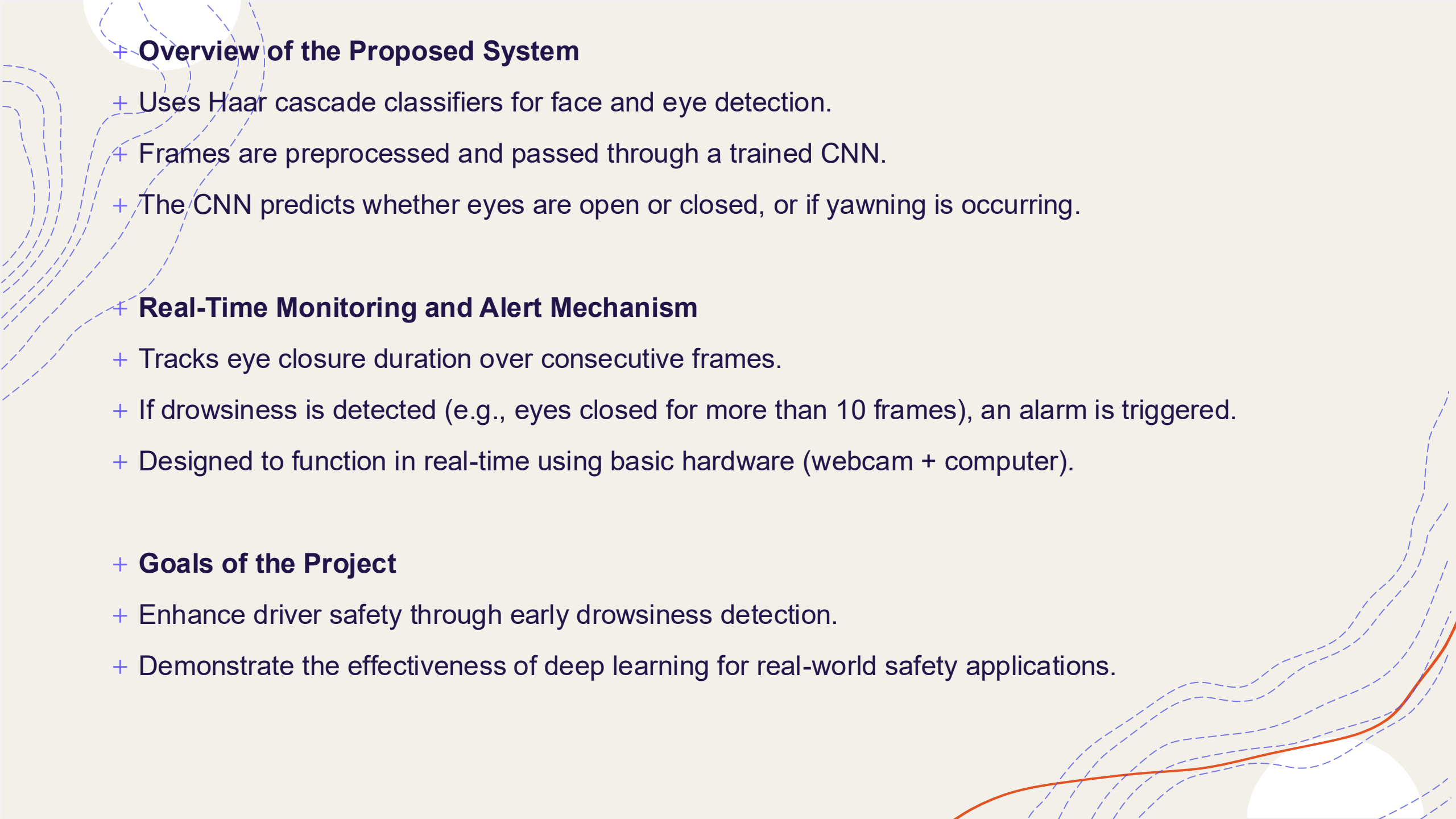
- + Driver drowsiness is a leading cause of road accidents globally.
- + Fatigue impairs reaction time, awareness, and decision-making.
- + Difficult to detect using traditional methods like steering pattern or lane tracking.

+ Need for Intelligent Detection

- + Conventional techniques are often inaccurate or require expensive external sensors.
- + There is a need for a **non-intrusive, cost-effective, real-time solution**.

+ Role of Computer Vision & Deep Learning

- + Computer vision enables real-time facial analysis using just a webcam.
- + **Convolutional Neural Networks (CNNs)** excel at detecting visual patterns like closed eyes and yawning.



+ **Overview of the Proposed System**

- + Uses Haar cascade classifiers for face and eye detection.
- + Frames are preprocessed and passed through a trained CNN.
- + The CNN predicts whether eyes are open or closed, or if yawning is occurring.

+ **Real-Time Monitoring and Alert Mechanism**

- + Tracks eye closure duration over consecutive frames.
- + If drowsiness is detected (e.g., eyes closed for more than 10 frames), an alarm is triggered.
- + Designed to function in real-time using basic hardware (webcam + computer).

+ **Goals of the Project**

- + Enhance driver safety through early drowsiness detection.
- + Demonstrate the effectiveness of deep learning for real-world safety applications.

Motivation

+ High Rate of Road Accidents Due to Fatigue

- + Drowsy driving is a major contributor to road accidents, especially on highways and long routes.
- + According to WHO and traffic safety reports, fatigue-related crashes often result in severe injuries or fatalities.

+ Need for Non-Invasive Detection Methods

- + Traditional detection systems rely on sensors or driver input, which can be intrusive or impractical.
- + There's a growing need for a **camera-based, real-time solution** that doesn't disturb the driver.

+ Rise in AI and Deep Learning Capabilities

- + Advances in computer vision and CNNs allow for accurate detection of subtle visual cues like eye closure, blinking rate, and yawning.
- + The reliability of these techniques motivates their use in life-critical systems.

Literature Survey

Sr no	Name of paper	Journal published and year	Author	Summary
1.	Driver Drowsiness Detection Model Using Convolutional Neural Networks Techniques for Android Application	International Conference on Communication, Computing and Virtualization	Rateb Jabbar, Mohammed Shinoy, Mohamed Kharbeche, Khalifa Al-Khalifa, Moez Krichen, Kamel Barkaoui	Approach: The authors propose a CNN-based drowsiness detection framework tailored for Android devices. It uses facial landmark extraction (via Dlib) to analyze eye movement and facial orientation, classifying drowsiness in real-time using images from the NTHU-DDD dataset. Innovation: Prioritizing computational efficiency, the model maintains a memory size of just 75 KB while achieving over 83% accuracy across multiple conditions (day/night, with/without glasses). The work bridges deep learning with mobile computing for accessible and scalable deployment.
2.	Driver Drowsiness Detection System Using CNN	International Journal of Science, Engineering and Technology (IJSET), Vol. 10, Issue 2, 2022	Soumya Agarwal, Dr. M.N. Nachappa	Approach: A CNN model is trained on a dataset of 30,000 images to detect driver fatigue through eye state analysis using webcam input. The system uses a multi-layer CNN architecture, with convolutional and pooling layers feeding into fully connected layers, classifying eye openness in real-time. Innovation: With an accuracy of 96–98%, the model provides timely audio alerts for closed-eye duration exceeding thresholds. It is designed for deployment on low-resource devices and emphasizes scalable safety applications for commercial vehicles and public transportation systems.

3.	Facial Region of Interest Based Driver Drowsiness Detection Using CNN Models	Applied Sciences (MDPI), 2023	Wilson Florez, Miguel A. Rodriguez, Jeysson H. Pedraza, Juan C. Fernandez, Carlos E. Montenegr	Approach: A comparative analysis of CNN architectures (InceptionV3, VGG16, ResNet50V2) is presented for eye-state classification using the NITYMED dataset, with MediaPipe utilized to precisely extract eye regions. The model detects drowsiness by analyzing blink duration and frequency in dynamic, real-world scenarios. Innovation: The research introduces a novel ROI correction method to maintain eye tracking accuracy during head movement. ResNet50V2 outperforms other architectures, achieving 99.71% accuracy, and visualization via Grad-CAM supports explainable AI in safety-critical applications.
4.	Driver Drowsiness Detection System Using CNN	International Journal of Research and Analytical Reviews (IJRAR), Vol. 8, Issue 3, July 2021	Sushmita Prasad, Nihal Shaikh, Trushita Jadhav, Robin Sen, Pushkar Sath	Approach: The authors developed a real-time driver drowsiness detection system using Convolutional Neural Networks (CNNs) to analyze eye closure over time. By processing video frames through OpenCV and TensorFlow, the system detects prolonged eye closure and triggers an alarm, effectively translating temporal visual features into fatigue cues. Innovation: They integrated boosting with weak classifiers (decision stumps) to iteratively enhance model accuracy, addressing common issues in traditional systems such as sensitivity to illumination changes. The model's lightweight design makes it suitable for deployment on Android and embedded systems.

5.	Drowsiness Detection in Real-Time via Convolutional Neural Networks and Transfer Learning	Journal of Engineering and Applied Science, 2024 (Vol. 71:122)	Dina Salem and Mohamed Waleed	Approach: This study applies transfer learning with pre-trained CNN architectures (e.g., InceptionV3) to detect drowsiness by classifying eye states (open, closed, yawning) using datasets like YAWDD and MRL. The model processes diverse visual cues from facial images and achieves high generalization through fine-tuning on domain-specific datasets. Innovation: The system achieves up to 99.86% accuracy in multi-class classification and includes a mobile application with a real-time alert system. Techniques like Grad-CAM are used to visualize feature attention, improving interpretability and model trustworthiness.
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Summary / finding of literature Survey

+ **CNNs as the Core Technique**

All papers employ CNNs for accurate eye state and yawning detection due to their effectiveness in image classification. Some use transfer learning to enhance performance.

+ **Real-Time Eye & Facial Monitoring**

Drowsiness detection relies on prolonged eye closure (typically >300ms), blinking patterns, and yawning to trigger alerts—enabling timely microsleep prevention.

+ **Diverse & Realistic Datasets**

Datasets like NTHU-DDD, YAWDD, CEW, and MRL Eye ensure robustness in real-world conditions (e.g., low light, head tilts), improving generalization.

+ **Lightweight & Mobile-Friendly Models**

Several models are optimized for Android and embedded systems, promoting accessibility beyond luxury vehicles to fleets and public transport.

+ **High Accuracy & Deployment Success**

Models report accuracies from 83% to 99%, with many implemented as real-time apps, bridging research with practical road safety applications.

Objectives

+ **Develop a Real-Time Drowsiness Detection System**

Design and implement a computer vision-based system that continuously monitors a driver's facial features to identify signs of fatigue or sleep.

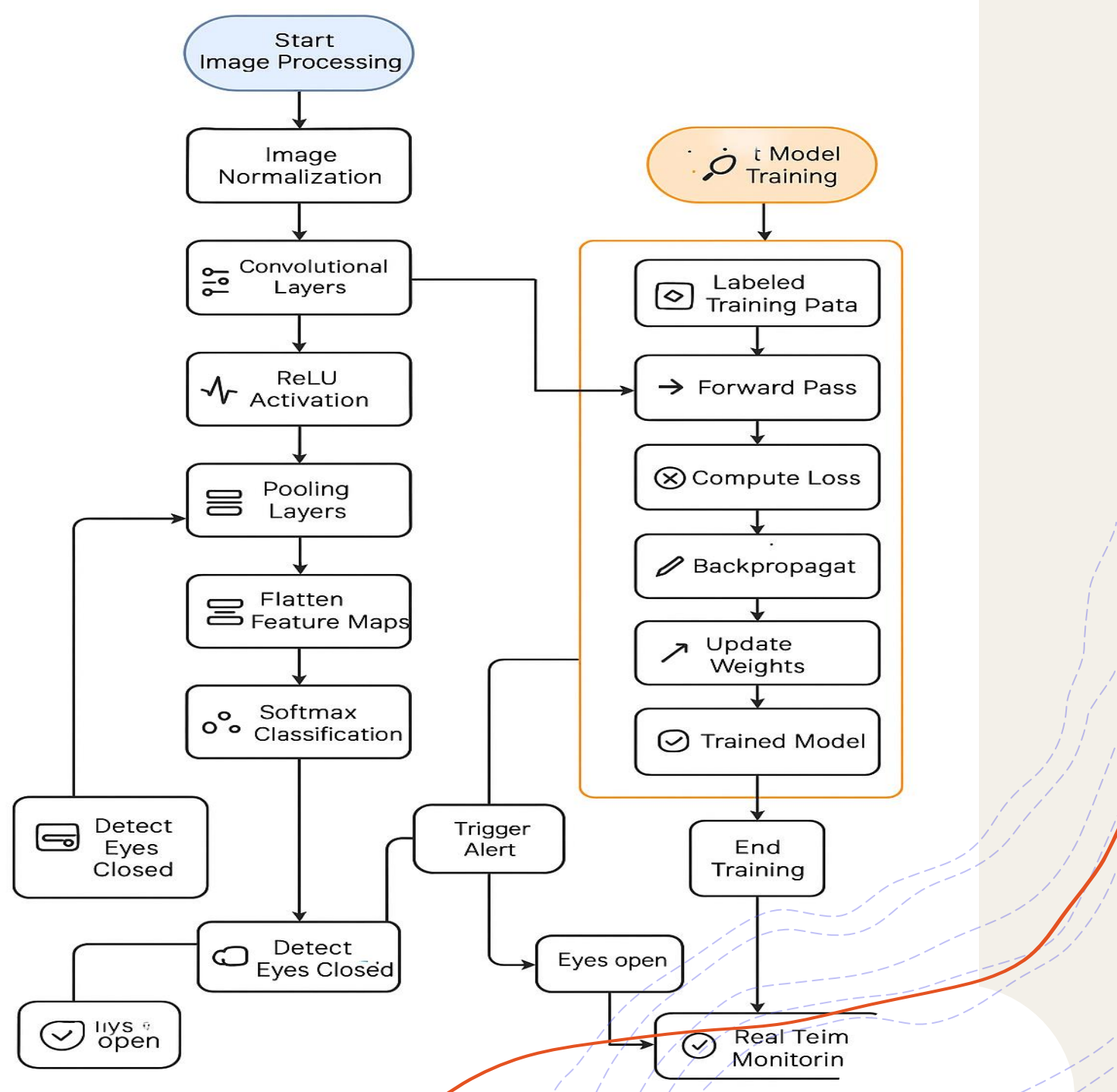
+ **Utilize Convolutional Neural Networks (CNNs) for Accurate Classification**

Leverage the power of CNNs to classify eye states (open/closed) and yawning with high precision using training data and live webcam input.

+ **Enhance Road Safety by Early Detection**

Trigger alerts when signs of drowsiness are detected for a certain period (eyes closed for 10+ frames), preventing accidents due to microsleeps

Architecture of System



+Image Processing Phase

- + **Start Image Processing:** The system begins by capturing video frames (typically through a webcam).
- + **Image Normalization:** Captured frames are preprocessed—resized (e.g., 145x145), converted to grayscale, and pixel values are normalized to improve CNN performance.
- + **Convolutional Layers:** These layers extract local patterns such as edges, eye contours, and facial landmarks from the input image.
- + **ReLU Activation:** A nonlinear activation function that introduces non-linearity into the model, helping to learn complex patterns.
- + **Pooling Layers:** Used for spatial reduction of feature maps (e.g., max pooling), reducing computation while retaining important features.
- + **Flatten Feature Maps:** Converts the pooled features into a 1D vector to feed into dense layers.
- + **Softmax Classification:** Final output layer gives probabilities of each class (e.g., eyes open, eyes closed, yawning).

+Model Training Block

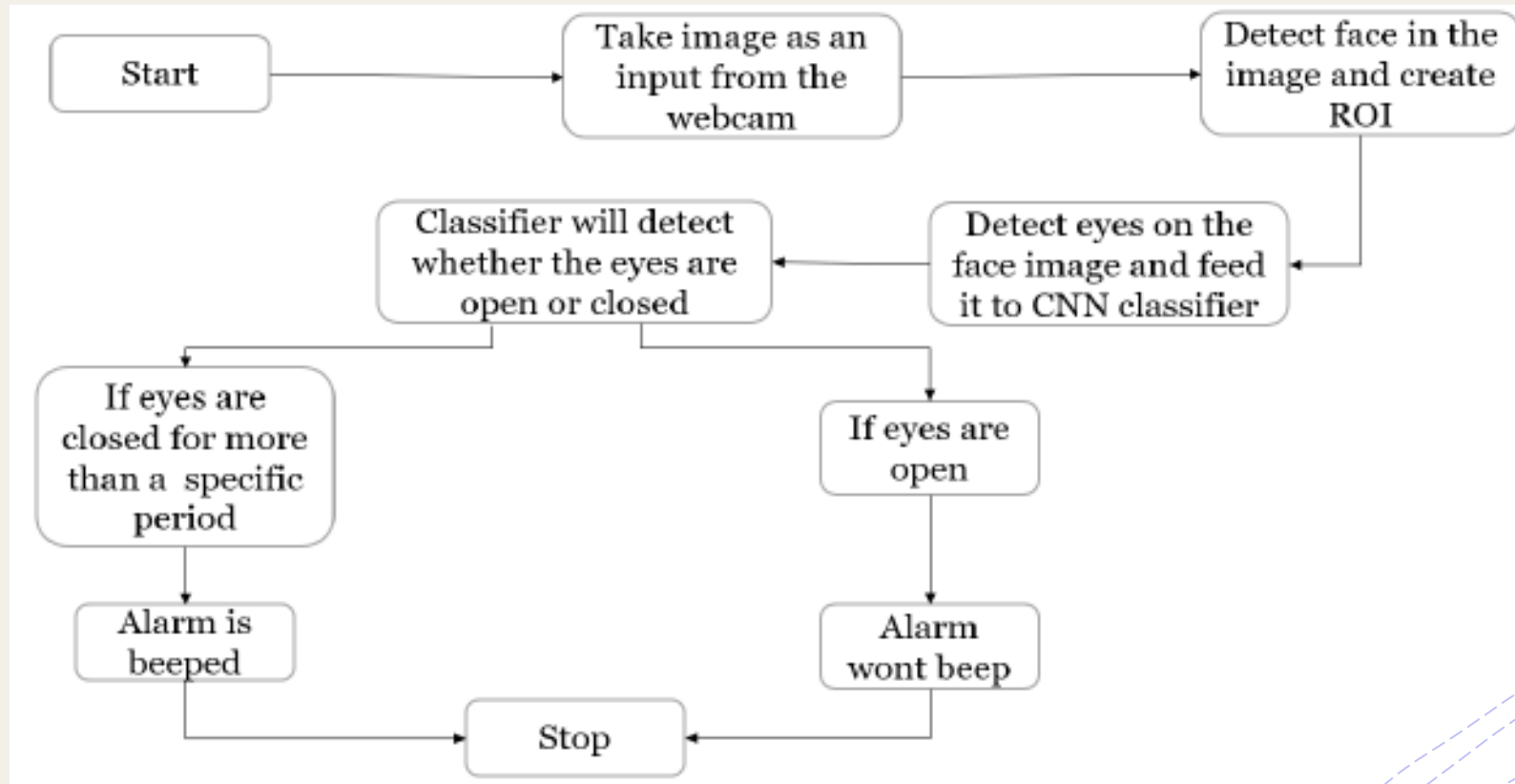
- + **Start Model Training:** Initiates the CNN training phase using labeled datasets.
- + **Labeled Training Data:** Images with annotated eye states (open/closed) are used for supervised training.
- + **Forward Pass:** Input data passes through the CNN model and outputs predictions.
- + **Compute Loss:** The difference between the predicted and actual values is calculated using a loss function (e.g., cross-entropy).

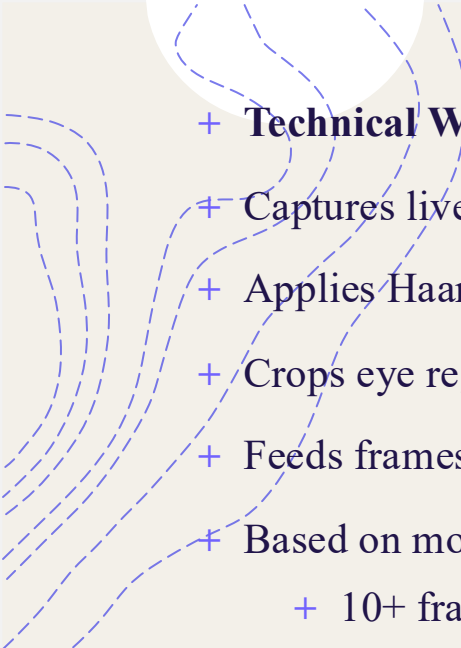
- + **Backpropagation:** Gradients are computed and passed backward to update the weights.
- + **Update Weights:** CNN parameters are adjusted to minimize the loss.
- + **Trained Model:** After multiple epochs, a trained model is produced.
- + **End Training:** Marks the completion of model training, ready for real-time deployment.

+3. Real-Time Monitoring Phase

- + **Prediction Output:** New frames are classified using the trained CNN.
- + **Detect Eyes Closed:** Logic checks if eyes are classified as "closed."
- + **Count Consecutive Frames:** If eyes remain closed for >10 frames, the system assumes drowsiness.
- + **Trigger Alert:** A visual/auditory alert is activated to wake the driver.
- + **Eyes Open Path:** If eyes open before the threshold, monitoring continues.
- + **End Real-Time Monitoring:** System ends session or continues based on user or external input.

Real time Monitoring





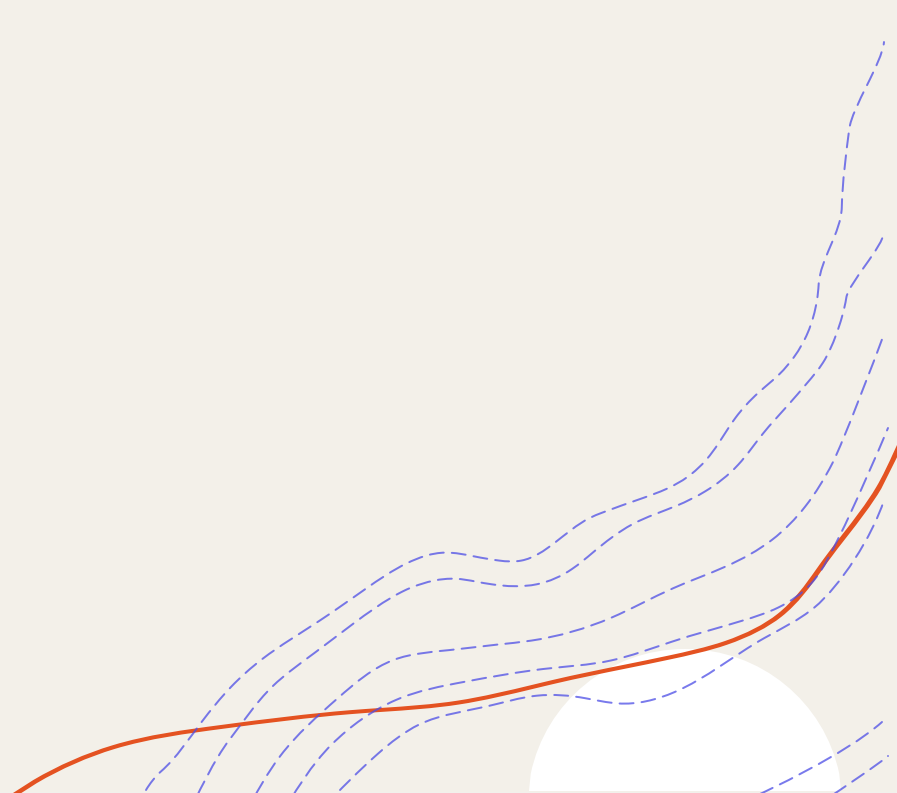
+ **Technical Workflow:**

- + Captures live video using OpenCV.
- + Applies Haar cascade detection for face and eye tracking in every frame.
- + Crops eye regions and resizes them.
- + Feeds frames into the trained CNN model.
- + Based on model output, triggers alerts after:
 - + 10+ frames of eyes closed.
 - + Yawning detected multiple times in short intervals.

+ **Output Mechanism**

- + Visual Alert: “Drowsiness Detected!” text on screen.
- + Auditory Alert: Buzzer or alert sound using playsound() or pygame.

+ **Advantages:**

- + Lightweight for edge devices.
 - + Works in varying lighting conditions due to augmented training.
 - + Scalable for commercial vehicle fleets, public transport, and personal cars.
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Implementation : CNN model

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 143, 143, 256)	7168
max_pooling2d (MaxPooling2D)	(None, 71, 71, 256)	0
conv2d_1 (Conv2D)	(None, 69, 69, 128)	295040
max_pooling2d_1 (MaxPooling2D)	(None, 34, 34, 128)	0
conv2d_2 (Conv2D)	(None, 32, 32, 64)	73792
max_pooling2d_2 (MaxPooling2D)	(None, 16, 16, 64)	0
conv2d_3 (Conv2D)	(None, 14, 14, 32)	18464
max_pooling2d_3 (MaxPooling2D)	(None, 7, 7, 32)	0
flatten (Flatten)	(None, 1568)	0
dropout (Dropout)	(None, 1568)	0
dense (Dense)	(None, 64)	100416
dense_1 (Dense)	(None, 4)	260
Total params: 495,140		
Trainable params: 495,140		
Non-trainable params: 0		

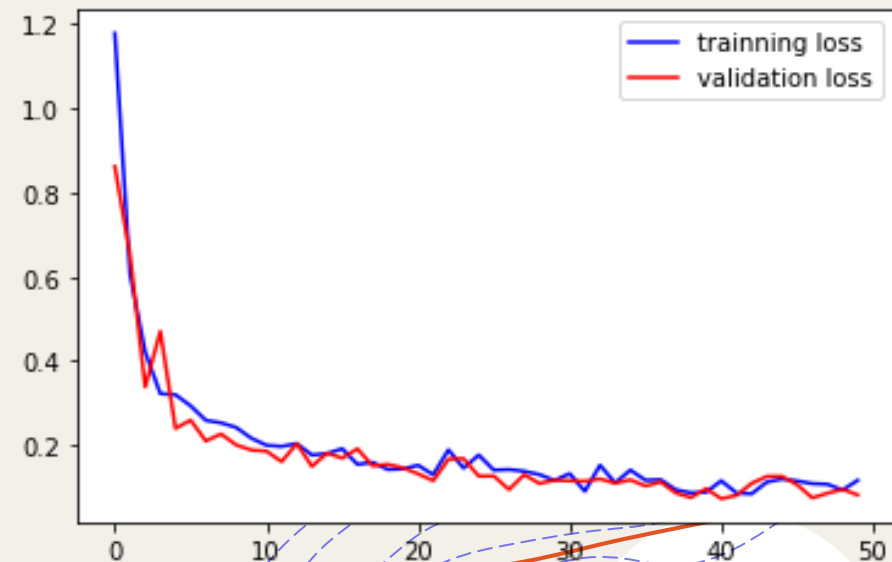
Training

```
Epoch 1/50
43/43 [=====] - 422s 10s/step - loss: 1.1787 - accuracy: 0.4595 - val_loss: 0.8622 - val_accuracy: 0.6747
Epoch 2/50
43/43 [=====] - 412s 10s/step - loss: 0.6066 - accuracy: 0.7506 - val_loss: 0.6524 - val_accuracy: 0.7180
Epoch 3/50
43/43 [=====] - 408s 9s/step - loss: 0.4214 - accuracy: 0.8411 - val_loss: 0.3382 - val_accuracy: 0.8668
Epoch 4/50
43/43 [=====] - 334s 8s/step - loss: 0.3222 - accuracy: 0.8656 - val_loss: 0.4703 - val_accuracy: 0.8270
Epoch 5/50
43/43 [=====] - 257s 6s/step - loss: 0.3199 - accuracy: 0.8708 - val_loss: 0.2397 - val_accuracy: 0.9048
Epoch 6/50
43/43 [=====] - 247s 6s/step - loss: 0.2934 - accuracy: 0.8775 - val_loss: 0.2595 - val_accuracy: 0.8858
Epoch 7/50
43/43 [=====] - 6473s 154s/step - loss: 0.2589 - accuracy: 0.9035 - val_loss: 0.2098 - val_accuracy: 0.90
Epoch 8/50
43/43 [=====] - 207s 5s/step - loss: 0.2530 - accuracy: 0.8990 - val_loss: 0.2265 - val_accuracy: 0.8979
Epoch 9/50
43/43 [=====] - 388s 9s/step - loss: 0.2421 - accuracy: 0.9072 - val_loss: 0.2006 - val_accuracy: 0.9048

Epoch 49/50
43/43 [=====] - 239s 6s/step - loss: 0.0934 - accuracy: 0.9710 - val_loss: 0.0949 - val_accuracy: 0.9706
Epoch 50/50
43/43 [=====] - 239s 6s/step - loss: 0.1165 - accuracy: 0.9562 - val_loss: 0.0817 - val_accuracy: 0.9619
```

Results

	precision	recall	f1-score	support
yawn	0.63	0.95	0.76	63
no_yawn	0.93	0.54	0.68	74
Closed	0.94	0.98	0.96	215
Open	0.98	0.94	0.96	226
accuracy			0.90	578
macro avg	0.87	0.85	0.84	578
weighted avg	0.92	0.90	0.90	578



Conclusion

- + Driver drowsiness detection using computer vision and deep learning techniques is an essential contribution to intelligent transportation systems and road safety. The project developed a real-time detection system that leverages convolutional neural networks (CNNs) to classify four distinct states—open eyes, closed eyes, yawning, and no yawning—based on facial features, specifically eye and mouth movements. The system was trained on a carefully preprocessed dataset, and a model was built that integrates Haar cascade classifiers for face and eye detection with a custom CNN for classification.
- + The achieved validation accuracy (~96%) highlights the model's ability to generalize on unseen data, demonstrating its suitability for practical scenarios. The system also successfully integrates alarm-based feedback to alert drowsy drivers, combining both software intelligence and user safety response. This hybrid design, involving classical image processing (for face/eye localization) and deep learning (for behavior classification), proved to be effective and modular.
- + However, despite its success, the model has certain limitations rooted in real-world unpredictability, including changes in lighting, facial occlusions, and subtle behavioral variations. The CNN, although effective, still relies on supervised training, which requires a robust and diverse dataset. Overall, the project not only showcases the technical feasibility of real-time drowsiness detection but also opens up opportunities for enhancement through multi-modal input and adaptive learning systems.

References

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